

Price Dynamics in the Counter-Strike 2 Skin Market

A Machine Learning Analysis using Median Daily
Price and 11 Years of Market Data

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Price Dynamics in the Counter-Strike 2 Skin Market: A Machine Learning Analysis using Median Daily Price and 11 Years of Market Data	
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This thesis investigates the short-term price dynamics of the Counter-Strike 2 (CS2) skin market and evaluates whether daily price movements can be predicted using machine-learning methods. A dataset of 640,145 daily observations covering 28 skins from 2013 to 2025 was constructed using data from the Steam Community Market. Feature engineering introduced economic, behavioural, temporal, event, and supply-related variables, including moving averages, price deviations, event indicators, and case-market metrics. Linear Regression, Random Forest, XGBoost, and an LSTM neural network were trained and evaluated using a chronological 80/20 split. Random Forest and XGBoost outperformed the linear model, achieving out-of-sample R^2 values near 0.50, while the LSTM model's performance was much weaker, indicating a lack of strong temporal dependency in price series. The most influential predictors were price-based features, particularly moving-average deviations, while esports events, Steam sales, and operations showed little influence on daily returns. Quantity sold was modestly affected by events, but prices were largely unaffected. The findings suggest that short-term skin price movements exhibit some nonlinear patterns but remain difficult to predict, consistent with a market driven by behavioural factors and micro-level noise rather than strong structural signals. The study highlights data limitations in virtual-goods markets and proposes that future work using individual-level transaction data and broader item coverage may improve predictive performance.	
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Prisdynamik på Counter-Strike 2 skins marknaden En maskininlärningsanalys baserad på daglig medianpris och elva års marknadsdata	
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Denna uppsats undersöker de kortsiktiga prisdynamikerna på marknaden för Counter-Strike 2-skins (CS2) och utvärderar huruvida dagliga prISRörelser kan förutsägas med hjälp av maskininlärningsmetoder. Ett datamaterial bestående av 640 145 dagliga observationer som omfattar 28 skins under perioden 2013–2025 konstruerades med data från Steam Community Market. Feature engineering introducerade ekonomiska, beteendemässiga, tidsmässiga, händelsebaserade och utbudsrelaterade variabler, inklusive glidande medelvärden, prisavvikelser, händelseindikatorer och mått relaterade till case-marknaden. Linjär regression, Random Forest, XGBoost och ett LSTM-neuralt nätverk tränades och utvärderades med en kronologisk 80/20-uppdelning. Random Forest och XGBoost presterade bättre än den linjära modellen och uppnådde R²-värden utanför stickprovet nära 0.50, medan LSTM-modellens prestanda var betydligt svagare, vilket indikerar avsaknad av stark tidsberoende struktur i prisserierna. De mest inflytelserika prediktorerna var prisbaserade variabler, särskilt avvikelser från glidande medelvärden, medan e-sportevenemang, Steam Sales och operationer hade begränsad påverkan på de dagliga avkastningarna. Antalet sålda enheter påverkades i viss mån av händelser, men priserna förblev i stort sett opåverkade. Resultaten tyder på att kortsiktiga prISRörelser för skins uppvisar vissa icke-linjära mönster men förblir svåra att förutsäga, vilket är förenligt med en marknad som i hög grad drivs av beteendemässiga faktorer och mikro-nivåbrus snarare än starka strukturella signaler. Studien belyser databegränsningar inom marknader för virtuella varor och föreslår att framtida forskning som använder transaktionsdata på individnivå samt ett bredare urval av objekt kan förbättra den prediktiva prestandan.	
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1. Introduction

The rise of virtual economies has transformed how value is created, exchanged, and perceived in digital environments. One of the most illustrative examples of this transformation is the marketplace for cosmetic in-game items, commonly referred to as skins, in the Counter-Strike series. These items, while lacking any impact on gameplay, have become highly sought-after digital commodities with real-world monetary value. Skins are often acquired through randomized loot boxes and can be traded or sold on digital marketplaces, giving rise to complex economies within and around the game ecosystem.

This thesis focuses on the skin economy of Counter-Strike 2 (CS2), the latest instalment in Valve Corporation's iconic first-person shooter franchise. Since the introduction of skins in 2013, in the previous title Counter-Strike: Global Offensive (CS:GO), the Counter-Strike skin market has evolved into a multi-billion-dollar ecosystem involving both developer-controlled and third-party marketplaces. These platforms enable the buying, selling, and trading of virtual items whose prices are influenced by a range of internal and external factors, including game updates, esports events, market trends, and community sentiment.

Despite the scale and visibility of these economies, there is limited academic understanding of what drives price fluctuations in skin markets, or how player behaviour and platform design contribute to the valuation of digital items. Furthermore, while price volatility and speculative trading are well-documented in traditional asset markets, it remains unclear how such dynamics operate in the hybrid space of virtual economies, where economic, cultural, and social logics intersect.

This thesis aims to address these gaps by applying machine learning (ML) methods to large-scale transaction and contextual data from the CS2 skin market. Specifically, it investigates whether internal (e.g., case drops, patch updates) and external (e.g., Steam sales, major esports tournaments) events can be used to explain and predict changes in skin prices in the short term.

1.1. Research Questions

- Which internal and external factors affect short- to medium term price movements in Counter-Strike skins?
- To what extent can machine learning models capture and predict price fluctuations in the Counter-Strike skin economy?

1.2. Purpose and Contribution

The primary purpose of this thesis is to analyse price dynamics in a virtual asset market using empirical and computational methods. It contributes to the literature on digital economies by combining theories of virtual goods and market behaviour with predictive analytics, offering insight into how valuation emerges in synthetic, platform-based marketplaces. Additionally, it bridges fields such as behavioural economics, platform studies, and applied machine learning by modelling player-driven market behaviour within a highly visible digital economy.

1.3. Thesis structure

The thesis is structured as follows:

- **Chapter 2** presents a review of the literature on virtual goods, speculative digital markets, and machine learning price prediction.
- **Chapter 3** presents context for the technical side of the skins and Steam Community Market.
- **Chapter 4** outlines the methodology, data collection, feature engineering, and model selection.
- **Chapter 5** describes the data, what parameters and tuning were done, and finally presents the results from the complete models.
- **Chapter 6** discussion of the findings, their implications, limitations of the study, and suggestions for future research.
- **Chapter 7** final conclusions about the thesis.

2. Literature review

This section examines the previous research done on virtual economies, value of skins, market behaviour, and machine learning price prediction.

2.1. Virtual economies and digital goods

Defining virtual goods can be complex due to their hybrid nature, existing somewhere between tangible property and digital services. Several authors have addressed this conceptual ambiguity of virtual goods. Fairfield (2005, p. 1053) focuses on their pseudo-tangible properties and argues that “virtual property is rivalrous, persistent, and interconnected code that mimics real world characteristics”. While authors, such as Lehdonvirta (2009) frame them as platform-

governed permissions. These permissions are governed by the platform's architecture and terms of service, which dictates how users can use, trade or sell items. Castranova (2001, 2005) adds an economic lens, emphasizing their real-world value. He notes that players in virtual environments attribute real world economic value to these items, even in the absence of legal ownership.

Castranova (2005) and Lehdonvirta and Ernkvist (2011) emphasize scarcity and how it affects the value and desirability of virtual goods. Castranova highlights scarcity as one of the factors contributing to the real-world economic value of virtual goods, while Lehdonvirta and Ernkvist argue that this scarcity is artificially controlled by the publishers of online games and social networking sites to make items desirable.

Virtual goods also vary significantly in type and function. Lehdonvirta and Hamari (2010) distinguish between functional virtual goods, decorative virtual goods, and social virtual goods. These classifications are not only aesthetic; they influence user behaviour, monetization strategies, and social dynamics within the platform. From a virtual economy design perspective virtual goods should contribute towards "(1) providing content, (2) attracting or retaining users, and/or (3) monetization" (Lehdonvirta & Castranova, 2014, p. 98).

Virtual economies refer to the systems of value exchange that develop around virtual goods, often within online games, social platforms, or virtual worlds (Lehdonvirta & Ernkvist 2011). Within games these economies often mimic real-world market dynamics, including supply and demand, inflation and labour specialization. While players work inside games to make or earn items or currency, those can then sometimes be sold or bought outside of the game for real money (Castranova, 2005). This has created whole industries of people farming in-game currency in games such as World of Warcraft and selling it for real money, even when it is against the terms of service of these games (Heeks, 2008). These markets can be considered secondary markets, which are marketplaces where players trade virtual goods among themselves for real money. Many games prohibit these forms of marketplaces and try to fight against them, while other developers create their own to monetize this trade between players (Lehdonvirta & Castranova, 2014)

Virtual economy marketplaces can generally be categorised into three types: in-world marketplaces, where virtual goods are traded only within the game environment; out-world marketplaces, which are developer-controlled platforms established outside the game; and third-party marketplaces, which emerge when no official marketplace exists and players create alternative channels for trading (Nazir & Lui, 2016).

One common way of selling virtual goods is through microtransactions. Free-to-play (F2P) games frequently use so called microtransactions, which are typically small purchases of premium content. This could include additional content, virtual currency or cosmetic items (Schwiddessen & Karius, 2018). Although microtransactions are generally small in monetary value, they are mainly determined by their purpose, and not by their size (Tomic, 2018). While microtransactions were initially used in F2P mobile games, they are now also found in some premium games that traditionally relied more on sales of the game itself (Schwiddessen & Karius, 2018).

One category of microtransactions is loot boxes, which are virtual items containing a randomised selection of other virtual items, ranging from cosmetics to gameplay-altering weapons, armour, or currency. Loot boxes typically have a collection of items with varying drop rates, meaning that some items are rarer than others (Schwiddessen & Karius, 2018). These types of microtransactions have been linked to gambling and have attracted increased scrutiny and criticism, leading Belgium to be the first country to ban them in 2018 (Denoo et al. 2023).

2.2. The nature of value in skin markets

Skins in CS2 offer no gameplay advantage; Thorhauge (2023) suggests that rather their value comes from the prestige of owning them attached to the skins by the community. Thorhauge further argues that because this, skins become suitable as assets and units of transaction. Another view is offered by Glaser (2022) who points out that self expression is an important factor in purchasing a virtual good. He also points out that the prestige, and therefore the value, can increase if professional players use a skin. In this way there is a paratextual narrative around skins, allowing you to show something like your wealth or affiliation through them.

Skins can be further analysed as what Latour (1993) would call *quasi-objects*, which are entities that blur the line between subject and object, material and immaterial, and the social and the technical. Fairfield (2022) supports this view, suggesting that virtual objects cannot be understood solely in technological terms, but must instead be seen as quasi-objects: simultaneously social and technical artefacts whose value emerges through interaction, community norms and platform policies. Thorhauge (2023) similarly argues that skins possess dual value: an in-game aesthetic value, and platform-level monetary value. This duality positions skins as hybrid artefacts that are both symbolic and tradable, embedded in fan culture while also functioning withing speculative markets.

The cultural value of skins also plays out in social spaces like Steam profiles and in-game lobbies. Werning (2019) notes that Steam allows players to use their collections for social

distinction, publicly displaying them to demonstrate taste, creativity, or wealth. In this sense, skins function similarly to fashion or collectible art, indicating a player's belonging within a community and potentially their seniority or status. These forms of expression support Thorhauge's (2023) observation that Steam addresses its users not only as players, but also as market participants, capable of shaping and responding to the cultural economy of skins.

While skins can hold cultural and social value, they have also been described as currency on their own. Thorhauge (2023, p. 117) "*Does this mean that skins represent a new type of game-based money and with a transactional community of gamers? Yes and no. Indeed, the vagueness of this answer may be the whole point.*" She analyses the skin market through Dodd's (1994) theory of "monetary networks", understood as networks of social relationships that make transactions possible. From this perspective skins can resemble a kind of currency, and Thorhauge notes that Valve has an interest in preventing them from being officially recognised as such. Although other scholars may not analyse it as deeply, many still refer to skins a currency, largely because they can be used as such on third-party sites. For example, Greer et al. (2019) describe skins as a virtual currency when examining the skins gambling that occurs on these external platforms.

Steam as a platform has been accused of "gamblification" by scholars like Macy and Hamari (2022) and Thorhauge (2021), referring to monetisation practices that incorporate elements of gambling. The gamblification occurs not only through the distributing virtual items via loot boxes but also through third-party sites explicitly designed for skin-based gambling. Zanescu et al. (2021) examined Steam's relationship to gambling and argued that the platform blurs the line between gaming and gambling. Although the study focused on Valve's other game *Dota 2*, the dynamics identified also apply to CS2; players engage in gambling-like behaviour in the hope of profit, whether through loot boxes or third-party gambling sites.

Holden and Erlich (2017) raised concerns about the legal aspect of third-party gambling and betting sites, where users can use their skins as wagers. Their critique emerged in the wake of multiple scandals involving match-fixing and high-profile streaming personalities gambling on sites without disclosing their affiliations. These controversies eventually lead Valve to release a statement in 2016, clarifying its stance on third-party gambling sites and banning their use through the Steam API and user agreement (Johnson, Valve 2016). However, Zanescu et al. (2021) argue that this ban remains selective in practice.

It is also important to note that Steam seeks to distance itself from being classified as a gambling entity by disallowing withdrawals directly through Steam. However, Zanescu et al. (2021) points out that this is only true when considering Steam in isolation. Within the broader

ecosystem of third-party sites, it is possible to convert skint into real-world currency, meaning that money does indeed exit the platform.

2.3. Investment and speculation

While symbolic and cultural value helps explain why players desire certain skins, there is also a speculative logic that shapes how players interact with the skin economy. In many respects, skins have evolved from mere cosmetics items into quasi-financial assets whose price movements are tracked and predicted on both official and third-party platforms.

Skin prices vary dramatically, ranging from a few cents at the lowest, to hundreds of thousands of dollars, depending on factors such as rarity, condition, pattern, and community demand. In 2024, for example, an *AK-47 Case Hardened* in Factory New condition reportedly sold for over USD 1 million, surpassing the previous record for the same skin in Minimal Wear condition for around USD 400 thousand in 2022 (Dinsdale, 2024). Data from the market aggregator CSGOSkins.gg (n.d.) illustrates this volatility: the *AWP Dragonlore* has increased by 4,200% between 2014 and 2025, rising from around USD 300 to around USD 13,000. Although not all skins have appreciated this sharply in price, or even appreciated at all, it does highlight the potential skins have as investments or vehicles for speculation.

Thorhauge (2021) argues that Steam positions its users not just as players, but also as “market speculators” or “stockbrokers”, actively engaging in value assessment and market behaviour. This speculative culture is reinforced by Valve’s platform design, which facilitates skin trading and selling while simultaneously controlling liquidity. The structure of these markets is shaped by platform affordances, that is, the ways in which Steam’s design encourages or constrains trading behaviour, as well as by algorithmic pricing mechanisms such as automated buy/sell orders and third-party price tracking tools that simulate financial trading environments. These tools contribute to a financialized atmosphere around skins, further embedding them in investment logics. Skin trading sites such as Pricempire provide portfolio-style views, market capitalisation charts, and tools for tracking individual skin price movements, closely mirroring the functionality of stock trading platforms.

Valve take a 15% cut of all sales of CS2 skin sales conducted on its own marketplace. Therefore, it could be argued that third-party function as competitors to Steam. However, both Zanesco et al. (2021) and Thorhauge (2023) contend that, although these third-party platforms draw traffic away from Steam, they essentially operate as complementors. According to Thorhauge, even though Valve does not profit directly from sales on external sites, their presence stimulates demand and contributes to overall growth of the skin economy.

Yamamoto and McArthur (2015) note that developers maintain full control over the game, meaning that any changes they implement can have significant effects on the in-game economy. They also observe that trade-up contracts, mechanisms in which ten lower-rarity skins can be exchanged for one of higher rarity, help stabilise the prices of lower-rarity items, even those that might otherwise be considered undesirable by the community. They conclude their study by suggesting that skin design, along with supply and demand dynamics, accounts for most of a skin's value.

2.4. Rational vs behavioural views of market dynamics

If skins can be considered investments, then it is important to understand market behaviour. In financial economics, two broad perspectives are often used to explain market dynamics: the rational view vs behavioural view. Understanding these can help us understand the behaviour of the skins market.

According to the Efficient Market Hypothesis (EMH), originally formalized by Eugene Fama (1970), markets are informationally efficient. This means that asset prices at any given moment reflect all available information, and that no investor can consistently achieve above-average returns without assuming above-average risk. In such market, opportunities for arbitrage are short-lived, as rational traders exploit them quickly, returning the market to equilibrium.

Behavioural Finance challenges the assumptions of perfect rationality by emphasising the psychological and emotional components of economic decision-making. Kahneman and Tversky (1979) developed Prospect Theory, demonstrating that individuals often make irrational choices, such as overweighting losses or following herd behaviour. This framework has been expanded by scholars like Barberis et al. (2001), who use it to explain stock market anomalies, and Thaler (2015), who examines how psychological biases such as loss aversion, mental accounting, and the endowment effect influence real-world economic and financial decision-making.

Although direct studies on the CS2 skin economy are limited, research on parallel markets such as NFTs price dynamics provide relevant insights. Barbon and Rinaldo (2023, preprint), in their study of NFT price dynamics, found that bubbles are both common and predictable. They distinguish between sophisticated investors, who tend to buy early and exit before a crash, and speculative actors, who are more susceptible to hype-driven behaviour. These findings suggest that market crashes are not anomalies, but the result of structural tendencies in speculative digital economies.

According to the EMH (Fama, 1970), markets remain efficient because there are enough rational traders to offset the effects of irrational ones. However, behavioural economists like Shleifer and Vishny (1997) argue that irrational traders can dominate market activity, particularly when rational arbitrage is risky, delayed, or costly. In such cases, mispricing can persist, and bubbles can form and burst in ways EMH cannot fully explain.

A central element underpinning market efficiency is information. EMH assumes that relevant information is freely available and quickly incorporated into prices. In the case of the CS skin market, this assumption is problematic. Valve does not disclose key information, such as when new skins will be released or what they will do with the game or skins in the future. This creates a low-transparency, high-uncertainty environment. As Grossman and Stiglitz (1980) argue, markets cannot be fully efficient when information is costly or unevenly distributed. Under such asymmetries exist, speculation often fills the gap.

Another key distinction between traditional asset markets and the CS skin economy lies in the motivations of buyers. In financial markets, investors typically aim to maximize returns based on available information and risk tolerance, an assumption foundational to EMH (Fama, 1970). By contrast, skin purchases are often motivated by aesthetic preference, social signalling, or cultural affiliation. As Thorhauge (2021) argues, skins in CS2 have both in-game cosmetic value and platform-specific monetary value, making them hybrid objects that serve expressive as well as speculative purposes. Players on Steam can define their own objectives beyond profit maximisation (Werning, 2019), such as purchasing skins that reflect their personal identity or convey social prestige. These non-monetary motivations complicate the notion of rationality and suggest that price formation in the skin market is influenced by symbolic, emotional, and community-based factors, an insight more aligned more with behavioural finance than classical models of market efficiency.

2.5. Machine learning in price prediction

Much has been written in recent years about the effectiveness of machine learning (ML) in modelling and predicting the behaviour of financial and speculative assets. While no peer-reviewed studies have been identified that apply ML directly to the pricing of CS2 skins, there is a growing body of literature demonstrating its value in related markets, including equities and cryptocurrencies.

A simple approach to studying virtual economies was presented by Kim et al. (2015), where they examine the relationship between forum sentiment and currency exchange rates in a massively multiplayer online game. Using lexicon-based sentiment analysis on large volumes

of player discussions, the authors show that shifts in aggregate sentiment are correlated with subsequent market movements, particularly downturns. Their findings suggest that player sentiment, as expressed through forums or social media, can act as a leading indicator in virtual economies.

In the stock market domain, researchers have explored a range of ML algorithms to capture nonlinear and temporal patterns in price data. A study by Fischer and Krauss (2018), for example, employed Long Short-Term Memory networks (LSTMs) to forecast the daily movements of the S&P 500. Their results showed that LSTM models significantly outperformed traditional logistic regression and deep neural networks (DNNs) in terms of directional accuracy, though Random Forests (RFs) occasionally offered comparable performance. These findings suggest that LSTMs are particularly effective in capturing sequential dependencies in financial time series data.

Further refinements have been made using variants such as Bidirectional LSTM (BI-LSTM), which processes input sequences both forward and backward. Sunny et al. (2020), for instance, found that BI-LSTM models outperformed unidirectional LSTMs in stock price forecasting, particularly when tuned with an optimal number of epochs and hidden layers. Their findings indicated that while deeper architectures can offer improvements, excessive complexity may reduce predictive accuracy; in their experiment, two hidden layers and 100 epochs produced the best results.

XGBoost (eXtreme Gradient Boosting) is another method frequently used in financial forecasting. Designed for performance and efficiency, it is well suited for structured data and data mining tasks due to its ability to handle missing data, prevent overfitting through regularisation, and provide interpretable feature importance. In stock market applications, Zhang (2023) employed XGBoost to forecast stock prices, achieving high predictive accuracy by incorporating a range of financial indicators such as moving averages and trading volume.

To further improve forecasting accuracy, some researchers have explored hybrid models that combine time series techniques with machine learning approaches. Wang and Guo (2019) proposed a hybrid model that integrates ARIMA with XGBoost to predict stock prices, demonstrating that the complementary strengths of these models can yield better performance than when used independently. Specifically, they introduced the DWT-ARIMA-GSXGB framework, which consists of three main components. First, the Discrete Wavelet Transform (DWT) is used to decompose the original time series into two components; a low-frequency approximation (capturing the trend) and a high-frequency residual (representing noise and nonlinear patterns). The ARIMA model is then applied to the approximation component to

model linear relationships, while the improved gradient-boosting model GSXGB (a variant of XGBoost) is used to model the nonlinear characteristics of the residual component. Finally, predictions from both components are recombined through wavelet reconstruction to produce the final forecast. Their results showed that this hybrid method outperformed traditional models in terms of prediction accuracy across ten stock datasets.

Machine learning has been increasingly applied to price prediction not only in financial markets such as equities and cryptocurrencies, but also in non-financial domains characterised by heterogeneous data structures. For example, Varshitha et al. (2022) conducted a comparative study on used car price prediction, evaluating several models including Deep Neural Networks, Random Forests, Linear Regression, LASSO, and Ridge Regression. Their results showed that Random Forests outperformed both linear and deep learning approaches. Similarly, Satapathy et al. (2022) compared a similar set of models, including XGBoost, and found that XGBoost achieved the highest predictive accuracy. Both studies operated with a relatively small number of input features (11 and 10, respectively) yet still demonstrated that machine learning algorithms can significantly outperform traditional regression-based approaches.

Feature engineering in ML is also a key factor in performance. Fayyaz et. al. (2025) used advance feature engineering techniques to improve price prediction in used car sales. They applied logarithmic transformation on price to reduce skewedness and improve model performance. Their models were trained using an 80/20 training–testing split, with Linear Regression serving as the baseline alongside machine learning models such as Random Forest and XGBoost.

Feature engineering has likewise been shown to improve performance in financial prediction tasks. Nabi et al. (2020) created additional variables to enhance their stock price prediction models, including the difference between the highest and lowest prices over the past month and the mean of opening and closing prices to capture average price movement. Models incorporating these engineered features achieved better overall performance than models without them.

Dinesh et al. (2021) further demonstrated the usefulness of feature engineering by employing moving averages to predict stock market price trends. Drawing on techniques from technical analysis, they used a shorter 10-day moving average and a longer 20-day moving average to capture short- and medium-term patterns. Similarly, for Bitcoin price prediction, Khedmati et al. (2020) incorporated moving averages into their machine learning models to enhance predictive accuracy.

3. Context

This chapter provides contextual background on CS2, including its case and skin systems, their technical foundations, and the broader infrastructure that supports them. It also outlines the role of Steam and the Steam Community Market, as well as event-based features relevant to the study. The description draws on both available literature and the author's own experience as a long-term player. The focus is limited to elements directly relevant to the research question, with the aim of clarifying how skins function, circulate, and interact within the game's ecosystem.

3.1. Counter-strike series

Counter-Strike is a long-running, team-based first-person shooter series developed by Valve. It is built around competitive, round-based matches in which two sides, the Terrorists and the Counter-Terrorists, compete to complete objectives or eliminate the opposing team.

Counter-Strike: Global Offensive, released in 2012, modernised the series with improved matchmaking and an expanded competitive ruleset. In 2023, the game was replaced by *Counter-Strike 2*, a full engine transition to the Source 2 platform. This update introduced enhanced lighting, volumetric smoke, reworked physics, and redesigned maps. Crucially for skin-related research, the migration to the new engine preserved players' existing inventories while updating materials and rendering techniques, resulting in visual changes that affected how some skins appear (Counter-Strike Official Site, n.d.).

3.2. What are skins

In CS2 (and its predecessor *Counter-Strike: Global Offensive*), skins are purely cosmetic items that change the appearance of in-game weapons, gloves, knives, and sometimes character models without altering gameplay mechanics.

Skins were first introduced to the franchise as part of the *Arms Deal* update in August 2013. According to CSGODatabase, more than 1,300 skins are currently listed for CS2 (2025).

Skins can be understood as customised variants of the base weapons used in the game. When a player owns a skin, they effectively possess an alternative version of the base weapon featuring a specific cosmetic design. This is reflected in the naming convention of skins, which combines the weapon name with the skin's designated finish (e.g., *AK-47 Slate* or *Desert Eagle Printstream*). Players can therefore own multiple versions of the same weapon type, each with different skins, and select which one to use during gameplay.

3.3. How are skins obtained and categorized

Players can acquire skins in several ways: through random drops for playing, by opening loot-boxes (“cases”) that require purchased keys to open, through trading, or via “trade-up contracts”, where multiple lower-rarity can be exchanged for higher-rarity one (cs.money, 2024).

Skins are organised into rarity tiers, each with a specific drop chance when obtained from cases (CSGOskins.gg, 2023):

- **Mil-Spec** (Blue): 79.92%
- **Restricted** (Purple): 15.98%
- **Classified** (Pink): 3.20%
- **Covert** (Red): 0.64%
- **Exceedingly Rare** (Knives/Gloves): 0.26%

Each dropped skin also has a 10% chance of being a StatTrak™ variant, which tracks in-game eliminations and features a visible counter on the weapon. (Pricempire, n.d.)

Skins are further defined by two randomly assigned technical attributes: wear rating and pattern template.

The wear rating is a decimal between 0.00 and 1.00 that determines the wear level or condition of a skin. Wear categories are distributed as follows (Totalcsgo, n.d.):

- **Factory New**: 0.00 – 0.07
- **Minimal Wear**: 0.07 – 0.15
- **Field-Tested**: 0.15 – 0.38
- **Well-Worn**: 0.38 – 0.45
- **Battle-Scarred**: 0.45 – 1.00

Wear ratings are immutable, meaning that they do not change over time, and they influence a skin’s visual appearance, with higher wear rating resulting in more scratches and wear. Wear ratings can also affect price within a wear category, as skins with values closer to the cutoff for the next wear category often command higher prices. For example, a Minimal Wear skin with a wear rating of 0.08 is typically valued higher than one with 0.14.

The pattern template, ranging from 1 to 999, controls the texture alignment of the skin. While it has no real effect on most skins, specific high-value items (e.g., Case Hardened series) depend heavily on pattern variation. The most coveted “Blue Gem” patterns, such as pattern

template 661 for the AK-47 | Case Hardened, are especially coveted and priced far above other variants (CSGOSkins.gg, 2024).

Stickers are another price-affecting feature. Although most add little value, rare or discontinued stickers, especially from early major tournaments, can significantly increase a skin's price. A skin can have up to five stickers on it.

For this study, it is important to note that wear ratings, pattern templates, and stickers are not included in the dataset. Due to limitations in the Steam API, these attributes are not available for historical sales. Nonetheless, they remain important factors that can influence individual skin prices.

3.4. Cases

Cases play a central role in shaping the supply of skins, while also functioning as market items with their own supply and demand dynamics. Although the methods for obtaining cases have evolved over the game's lifetime, they generally follow a system of random drops obtained through gameplay. Under the current system, players can earn random skins and cases through a “weekly care package”, which is awarded once per week through regular play (csgodatabase.com, n.d.).

Cases can be categorised into three groups: active drop pool cases, rare cases, and discontinued cases. The active drop pool consists of the most recently released cases and represents the most common source of case drops. Older cases transition into the rare category, where they have a substantially lower probability of dropping. Discontinued cases are no longer obtainable through gameplay and therefore have no drop chance at all (csgodatabase.com, n.d.).

Like skins, cases can be traded or sold on the Steam Community Market and have independent prices and supply. Older and discontinued cases typically exhibit lower supply and higher market prices.

3.5. How are skins stored and traded

Skins and cases in CS2 (and CS:GO) are stored as digital inventory items linked to a player's account on Steam. This means that skins are centralized on Steam's servers. Each skin exists as an entry in the user's Steam Inventory, where it is associated with unique metadata such as item ID, finish type, wear rating, pattern template, and, when applicable, StatTrak data. This structure allows skins to function as discrete virtual assets that persist independently. (Steamworks, n.d.)

Trading and selling occur through two main mechanisms: the Steam Community Market and peer-to-peer item trades. The Steam Community Market facilitates sales between users through a centralised marketplace operated by Valve, allowing players to list items at chosen prices and purchase skins using funds stored in their Steam Wallet. Peer-to-peer trading, by contrast, enables users to exchange items directly through Steam's built-in trade interface.

3.6. Steam Community Market

This functions as a marketplace for trading skins between players. It allows a person to list a skin on the marketplace for a price between USD 1,800 the highest and 3 cents at the lowest, or similar in other currencies. Skins on the community market are bought using funds on the Steam platform with the wallet function. This allows people to store up to USD 2,000 and use it freely on the platform. While funds can be added to your account, importantly they cannot be withdrawn. The limits on sales and wallet funds have changed over time and might change in the future (Steam Support FAQ, n.d.).

Players can freely look over the skins on sale, as well as inspect them in-game to get a better view of the skin and its technical aspects like wear rating and pattern template. They can then choose which one to buy or place a buy order for a certain price. When a skin is listed on the market for the price of the buy order, the sale of that skin is automatically made. Both buyers and sellers can see what current buy orders exist to either offer more as a buyer, or to instantly sell a skin for the highest buy order price.

The Steam Community Market takes a cut for each sold item on the market, this consists of a "Steam Transaction Fee" of 5%, which applies to any game using the Steam Community Market, as well as a game specific fee of 10% in the case of CS2 (Steam Support FAQ, n.d.). This means that the total cut for each sale is 15%.

3.7. Third-party marketplaces and trading outside of the Steam Community Market

Third-party marketplaces are not involved in this study; however, they play an impactful role on the skin's economy. They operate with their own rules and limitations, with the most important aspects being that they allow withdrawing of funds to real currencies, and in general allow sales outside of the price limit of USD 1,800 set by the Steam Community Market.

This is possible through the use of the peer-to-peer trading function between different Steam users. By this way third-party sites can facilitate trades between players, allowing players to set and negotiate prices between themselves and the site handling the transaction of money

and skin between the seller and buyer. Some third-party sites also have their own inventory of skins and set prices for selling and buying skins.

Trading skins outside of the Steam Community Market comes with its own risk, particularly scams (CSGOSkins, 2023). Once a trade is made, there is no way of reverting it and Steam will rarely if ever help recover lost items (Khan, 2025). For this reason, some might avoid using third-party sites, even if they offer better fees and possibility of converting skins to real currencies.

3.8. Events

This section describes the major in-game and external events incorporated into the dataset. These events were selected because they may influence player activity, market behaviour, and short-term fluctuations in skin prices.

3.8.1. Steam sales

The Steam platform is mainly a platform for users to buy and store digital versions of games. While discounts on games sold on Steam can happen at any point, the so-called Steam Sales function as major discounts on multiple games at once. These happen periodically over the year, with Winter, Summer, Autumn and Fall sales (Betts, 2025). While skins themselves have no discounts, people can sell their skins on the community market and use the funds to purchase games.

3.8.2. Majors

These are Valve sponsored tournaments and generally have the highest prestige for winning them. The way the function has changed over time, but they usually involve:

- Capsules, sold by Valve, with random player and team stickers from participating teams.
- Pick'em challenge for predicting teams advancing to the next stage of the tournament.
- In-game advertising for the tournament.
- Drops of souvenir skins, which are skins only dropped when Major tournaments are ongoing and have unique player and team stickers on them.

(Bozhenko, 2025)

Majors could therefore lead to more interest in the game, and skins as viewers of the games get to see professional players use many different skins, often highly expensive ones.

3.8.3. S-tier tournaments

These are tournaments with higher price pools, usually played offline and have the best teams participating. These tournaments usually hold the most prestige for viewers and players and are held by well-established organizers (Liquipedia, n.d.). While not being as prestigious as Majors, some S-tier tournaments still reach high viewership.

3.8.4. Operations

Operations a form of purchasable DLC (Downloadable Content), which offers extra content and lasts around two to three months. The way operations work can change from one to another, but generally have both exclusive and community made content in the form of new maps and missions (Liquipedia, n.d.). With more interest and new content skins might be affected by these events.

4. Methodology

This section provides an overview of the data-gathering procedure, the feature-engineering steps, the model-selection, and evaluation of the models used in the study.

4.1. Data gathering

The primary dataset used in this study consists of transaction information from the Steam Community Market, obtained through Valve’s official API. The API provides aggregated daily sales data for a specific skin, including their condition and StatTrak status. Data collection was automated using a Python script, which searches for a skin and converted the JSON string to an Excel table that was subsequently enriched with additional variables.

For each day, the dataset reports the median sale price and the number of items sold. More detailed transaction-level data (such as individual sale prices or buyer information) are not publicly accessible through this interface, limiting the granularity of the analysis. While relying on median daily prices presents a constraint, it simultaneously reduces the influence of extreme outliers on the dataset.

To maintain temporal consistency, the most recent 30 days of data were excluded, as Steam’s API reports these days with higher frequency and partial updates, which could introduce bias in daily aggregation.

In addition to skin-level data, corresponding case sales data were retrieved through the same API, following the same structure of daily median price and quantity sold. This data was later linked to the relevant skins based on their associated cases.

To enrich the market data with contextual variables, several secondary sources were incorporated:

- Player activity data with the daily peak player counts were scraped from SteamDB. SteamDB also provides timing on of major Steam platform sales used in this study.
- Case release dates and Operation periods (start and end dates) were compiled from verified community and archival sources documenting content updates.
- Esports tournament data, including all S-tier events and Majors, were obtained from Liquipedia, which provides detailed information on tournament schedules and prize pools.

All secondary sources were integrated into the main dataset using the date variable as the key, allowing daily market activity to be aligned with relevant in-game and external events.

The data obtained directly from the Steam API can be considered highly reliable, as they originate from Valve's official transaction records. Secondary sources such as SteamDB may contain minor discrepancies, particularly for historical player count figures; however, these differences are small and do not materially affect long-term trends. Event-date information (including Majors, S-tier tournaments, Steam Sales, and Operations) is generally reliable, as it is consistently documented across multiple independent community and archival sources. Cross-validation revealed no substantial inconsistencies.

Game updates were left out of the study; this is because a lot of them are very minor. CS2 is relatively stable game, where rather than having game changing updates, the updates are typically very minor with small changes to maps or performance. Usually, bigger updates come with a new case, which is already tracked through dates for new case releases.

4.2. Skin selection

Given that the total number of available weapon skins in CS2 exceeds 1,300 as of 2025, including the full population was neither feasible nor necessary for this study. To ensure manageability and computational efficiency, a purposive sample of 28 skins was selected. This includes all conditions available and StatTrak versions of each skin.

The sampling strategy aimed to capture a diverse cross-section of market characteristics that are likely to influence price behaviour. Selection criteria included price range, release date, weapon popularity, and rarity. The objective was not to construct a statistically random sample,

but to ensure representation across different market segments and item attributes relevant to the analysis.

The first step in the selection process was to limit the scope to the seven most used weapons in competitive gameplay; AK-47, M4A1-S, M4A4, AWP, Glock-18, USP-S, and Desert Eagle. These weapons represent the most used guns in gameplay and therefore should have more interest from players. Four skins for each gun were selected for a total of 28.

Price was a central consideration. Extremely high-value skins often trade infrequently or are constrained by Steam's market price cap of USD 1,800. To maintain sufficient data density, the focus was placed on skins typically valued between approximately EUR 1 and EUR 100, a range associated with regular market activity. Because skin prices vary by wear condition and over time, the boundaries of this range were treated flexibly.

Additional emphasis was placed on diversity of release dates and rarities. Including skins from different cases and time periods allows for analysis of how supply age and rarity affect price behaviour. Skins released within the last two years were excluded to avoid insufficiently established markets and incomplete historical data.

The specific skins were hand-selected using information from CS2Goskins.gg, which lists all skins for each weapon along with their popularity, release date, and rarity. For each weapon, the most popular skins were reviewed and chosen to maximize diversity across rarities and release periods. Skins that are not obtainable through cases were excluded, ensuring consistency in market availability mechanisms.

Although the resulting set does not represent a perfectly balanced distribution of rarities and release dates, it was considered sufficiently diverse to capture the relevant range of market behaviours for this analysis.

Knives and gloves were deliberately excluded from this study. These items represent some of the rarest and most expensive assets in the CS2 market, with a drop chance of only 0.26%. Their market dynamics differ substantially from case-drop weapon skins. Many knife and glove finishes exhibit significant variation across individual pattern templates, which could create large price differences that cannot be captured through the aggregate daily median prices provided by the Steam API. Furthermore, transaction histories show that high-value items often sell infrequently, sometimes with weeks between sales, resulting in extremely sparse and irregular time series. Together, these characteristics introduce additional complexities that would require specialised modelling and feature engineering.

All skins chosen can be found in StatTrak and while most skins can be found in all the different conditions, some skins have a smaller than normal wear rating range resulting in

certain conditions being unavailable. Table 1 shows the distribution of Classification of the skins chosen. The skins chosen are from 22 different cases, and the maximum number of skins from a single collection is two.

Full list of skins chosen in appendix A.

Table 1 Classification count

Classification	Count
Mil-Spec Gade	7
Restricted	5
Classified	6
Covert	10

4.3. Feature engineering

The purpose of the feature engineering process in this study was to transform raw transactional and contextual data into variables that capture short-term market dynamics in the CS2 skin economy.

While long-run structural trends are present in both player activity and skin prices, the primary interest of this analysis lies in short term fluctuations, such as daily returns, event-driven changes, and week-to-week market adjustments. Therefore, the engineered variables emphasize transformations that stabilize variance, remove long-term trends, and highlight temporal patterns relevant for explaining short-term price movements.

Principles for data transformations come from traditional econometrics and consist of mathematical transformations. Hyndman and Athanasopoulos (2018) suggest using natural logarithmic transformations to stabilize variance and moving averages as a smoothing method to reveal underlying trends. Tsay (2010) suggests using log returns for timeseries forecasting because it is more tractable statistically. Log returns are closer to stationary and have more stable variance. Models that have non-stationary variables can lead to spurious regression where statistically meaningful coefficients are found even when a meaningful relationship does not exist (Harris & Sollis, 2003).

For these reasons, rather than using the raw median price, the target variable in this study is defined as the one-day logarithmic return. In addition, moving averages are employed as a smoothing technique, and simple percentage changes are used to capture short-term variation between consecutive observations. These transformations are applied in a manner

appropriate to each variable type (e.g., price, quantity, or peak player counts) to emphasise the most relevant temporal dynamics in each feature.

More specific transformations for each feature are described in the following sections.

4.3.1. *Skin related variables*

Each observation contains identifiers describing the skin and its characteristics. These include the skin name, its StatTrak status, the weapon type (e.g., AK-47, M4A1-S, Glock-18), and the rarity classification (Covert, Classified, Restricted, Mil-Spec). Weapon type and rarity are categorical variables and were encoded as dummy variables for modelling purposes.

An additional temporal variable, `days_since_skin_release`, was created to represent the number of days since the skin was first introduced to the game. This was calculated based on each item's official release date rather than its first recorded sale.

These part of the preprocessing was performed in Excel before the dataset was imported into Python. Basic attributes were added for each skin manually using simple spreadsheet functions. Days since release counter was computed in Excel by subtracting the skin's release date from each observation date.

4.3.2. *Price-related variables*

Skin prices exhibit substantial variation both across items and over time. To stabilize variance and improve comparability, the daily median price was transformed using the natural logarithm, producing the variable `price_log`.

From median price and `price_log`, several derived variables were created to capture short- and medium-term market dynamics:

- `Price_logret_1d`: daily logarithmic percentage change, representing short-term change. (Target variable)
- `Price_logret_7d`: seven-day logarithmic return, representing medium-term trends.
- `Price_ma7`: seven-day moving average, indicating general price direction.
- `Price_deviation_ma7`: deviation of today's log price from seven-day moving average, capturing relative over- or undervaluation.
- `Price_tier`: categorical grouping of skins based on long-run average price (0–10 = cheap; 10–50 = mid-range; 50–100 = high-end; 100+ = luxury).

These variables collectively capture price levels, momentum, and valuation dynamics across skins.

4.3.3. Volume-related variables

Transaction volume reflects market demand and liquidity. The raw number of daily sales was log-transformed to produce `log_quantity`, reducing skewness and aligning the scale with the price variables.

Two additional variables describe temporal patterns in demand:

- `Quantity_ma7`: seven-day moving average, capturing medium-term demand trends.
- `Quantity_pct_change`: day-to-day percentage change, representing short-term volatility in sales activity.

4.3.4. Case-related variables

Case-related variables were included to represent supply-side conditions in the skin market.

Three engineered features were constructed:

- `Case_quantity_ma7`: seven-day moving average of daily case sales.
- `Case_quantity_log`: logarithmic transformation of case sales, stabilizing variance and compressing large spikes.
- `Case_price_LogMA7`: seven-day moving average of logarithmic case prices, capturing medium-term trends in case value.

These variables provide information about case-opening activity and underlying supply mechanisms. As with skin prices, the raw case price and raw quantity variables were not used directly in modelling.

During preprocessing, approximately 6,000 missing values (about 0.9% of observations) were observed in case-level variables. Since the Steam Community Market does not report days with zero sales, and because gaps can also arise from minor inconsistencies in API coverage or key matching, the reasonable and conservative assumption is that a missing value reflects no reported transactions for that day. Following common practice in time series econometrics, missing case prices were forward filled to preserve price continuity, and missing case quantities were set to zero, reflecting the absence of sales.

After these adjustments, 407 observations still lacked valid case price information. These cases likely represent unmatched dates. Because they constitute less than 0.1% of the dataset and cannot be reliably imputed without introducing artificial values, these remaining rows were removed prior to modelling. This ensures that the modelling dataset contains only fully defined feature vectors, avoiding distortions in learning algorithms.

4.3.5. *Peak player count*

The daily peak player count reflects overall engagement with the game but exhibits strong long-run growth and clear non-stationarity across the 11-year period. Missing peak player values were forward-filled (and backward-filled for initial gaps) to create a continuous series for computing logarithmic transformations, percentage changes, and moving averages. The raw values were not used as model inputs.

To capture meaningful short- and medium-term variation, player numbers were transformed into:

- Log_peak_players: logarithmic scale to stabilize variance.
- Player_pct_change: daily percentage change, capturing short-term shifts in player activity.
- Player_ma7: seven-day moving average, capturing medium-term trends.

These transformations allow the models to detect event-driven fluctuations in player engagement rather than long-run structural trends. The raw player count variable was therefore excluded from modelling.

4.3.6. *Temporal and event-related variables*

To capture recurring patterns, the weekday of each observation was recorded (Weekday). This was done using a standard Excel formula and was added initial preprocessing together with skin-related variables.

Event-related features were matched directly to dates and encoded as binary indicators (1 = ongoing, 0 = not ongoing). For Majors and Cases counters were introduced, these count the days since the last Major ended, and the latest case was released.

Although Majors are a subset of S-tier tournaments, they were modelled separately. To avoid multicollinearity, Majors were removed from the S-tier list. Price pool information for tournaments was also added initially; however, it was removed prior to modelling because it exhibits no meaningful daily variation and lacks theoretical relevance for short-term price movements. Including it could have introduced unnecessary noise into the data without contributing any meaningful predictive value.

4.3.7. *Summary*

The final dataset integrates intrinsic item characteristics, price and volume dynamics, player engagement data, and context-specific event information. This comprehensive feature set 22

enables the subsequent models to analyse how both internal market behaviour and external events influence CS2 skin prices, especially over short periods.

Before the modelling, any rows with missing values are removed. This includes calculations that require a set number of days to calculate (e.g. moving average). For this reason, the first seven days of a skins market data are automatically removed from the final dataset. This is not considered an issue, because the prices are highly volatile the first few days.

4.4. Model selection

The aim of the modelling stage is to explain and predict short-term changes in skin prices using the engineered features described above. Because the market exhibits nonlinear relationships, potential event-driven shocks, and sequential dependence, a combination of linear, tree-based, and neural network models was selected. Using multiple models allows for both interpretability and predictive benchmarking. Models were selected based on the literature review process with similar studies regarding financial assets, cryptocurrencies and online marketplaces. The final selection of models are:

- Linear Regression as the baseline,
- Random Forest Regression,
- XGBoost, and
- Long Short-Term Memory (LSTM) network.

4.4.1. Modelling objectives and requirements

The primary objectives of the models are:

1. To identify which features most strongly explain short-term price movements,
2. To evaluate how external events influence market behaviour, and
3. To compare predictive performance across different modelling approaches.

Because of the dataset constructed and the target being short-term price prediction the model should be able to:

- Handle nonlinear data,
- Handle many features,
- Capture temporal dependence,
- Handle short-term predictions, and
- Allow for interpretability.

Table 2 describes how the different models handle these requirements. Linear regression does not meet many of the requirements, but still serves as a reference for the other nonlinear models.

The weakness of XGBoost and RF models is that they cannot capture temporal dependencies on their own, but with the feature engineering transformations done in this study (e.g., moving average, log returns) allows the models to capture some temporal dependencies.

Interpretability will make it possible to determine which features contribute to price movements. However, due to the nature of deep learning models, interpretability for the LSTM model is limited and not pursued in this study.

Table 2 Model requirements

Requirement	Linear Regression	Random Forest	XGBoost	LSTM
Handle nonlinear data	No	Yes	Yes	Yes
Handle many features	Yes	Yes	Yes	Yes
Capture temporal dependence	No	No	No	Yes
Short-horizon prediction	No	Yes	Yes	Yes
Feature interpretability	Yes	Yes	Yes	No

4.4.2. Linear regression

Linear regression is used as a baseline because it provides interpretable coefficients and allows for direct examination of the relationship between individual features, such as event indicators or player activity, and price movements. Although it cannot capture nonlinear dynamics, it establishes a benchmark against which all more advanced models are compared.

4.4.3. *Random Forest Regression*

Random Forests can capture nonlinear relationships and feature interactions that linear models cannot. The algorithm constructs multiple unique decision trees using bootstrap samples and random feature subsets, which reduces variance and improves generalisation. For regression tasks, the final prediction is obtained by averaging the outputs of all trees (Kuhn & Johnson, 2013). Random Forest models allow for interpretability by showing feature importance.

4.4.4. *Gradient Boosting Regression (XGBoost)*

Boosting models are included due to their strong empirical performance in complex regression tasks (Chen & Guestrin, 2016). XGBoost is a scalable tree boosting system with high performance, flexibility and robustness. It is well suited for tabular data with a mix of numeric and categorical features and can handle large datasets. Importantly for this study, XGBoost provides feature importance scores that allow for interpretability (XGBoosting, n.d.).

4.4.5. *Long Short-Term Memory network (LSTM)*

LSTM networks (Hochreiter & Schmidhuber, 1997) are specifically designed for sequential data and provide a mechanism for learning temporal dependencies through gated memory cells. Unlike other models used in this study, LSTMs can capture lagged relationships on their own. This includes short-term momentum patterns and nonlinear temporal structures. For this reason, LSTMs complement the static models above. LSTMs are a form of deep learning and have been used for predicting stocks in previous studies (e.g., Fischer & Krauss, 2018).

4.4.6. *Hyperparameter search*

Although the primary goal for this study is not exhaustively optimizing model performance, limited hyperparameter searches were performed for Random Forest and XGBoost models.

This was done to:

1. Verify that the baseline configurations were not severely underperforming.
2. Whether model adjustments could yield meaningful improvements to accuracy.

Given the datasets size, exhaustive grid search was avoided in favour of lightweight exploratory tuning.

4.4.7. *Evaluation strategy*

All models were built using a chronological train–test split to respect the temporal ordering of the data and prevent information leakage from the future into the past. The dataset exhibits strong temporal structure, with skins being introduced gradually over the eleven-year period

and certain event types (such as Operations and Majors) occurring more frequently in earlier years. Consequently, the distribution of event variables is not uniform across time, and some events are underrepresented in the later years of the dataset.

Despite this, a chronological train–test split was used, as it is the only method that respects the time-series nature of the problem and prevents look-ahead bias. Random splitting was avoided because it would mix future observations into the training period and produce overly optimistic results. The implications of uneven event frequency are discussed in the results, but they do not invalidate the forecasting setup; instead, they highlight that some event features may have limited predictive power in recent years.

Predictive performance was evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) to quantify average forecast errors, along with the coefficient of determination (R^2) to assess how much of the variance the model can explain. These metrics are widely used in econometrics and machine learning for evaluating regression-based forecasting models.

4.5. Use of AI in this study

During the writing process, ChatGPT was used primarily to assist with language refinement and to identify inconsistencies in academic writing. In addition, it was occasionally used to support the structuring of the thesis, suggest relevant literature, and assist with the identification and correction of minor coding errors.

5. Results

This section presents empirical results of the study. It begins with a descriptive overview of the dataset and then examines each predictive model in turn. Model accuracy is evaluated using MAE, RMSE, and R^2 , and patterns in feature importance are also analysed.

A short quantity analysis is also performed to see whether events impact the quantity of skins sold. This is accomplished through a simple linear regression and comparing sales averages during and off events

5.1. Descriptive statistics

After preprocessing and the removal of 7,808 observations containing missing values, the final modelling dataset contains 640,145 daily observations and 49 variables. The data span the period from 28 November 2013 to 31 January 2025, covering more than eleven years of activity on the Steam Community Market. The full description of each variable in the dataset can be found in Appendix B.

Across all selected skins, the median daily sale price has a mean of EUR 35.77, ranging from EUR 0.04 to EUR 1,721.74. Daily transaction volume also varies substantially, with an average of 184.75 items sold, a minimum of 1, and a maximum of 4,102.

The distribution of daily log returns (Figure 1) shows a strong concentration around zero, indicating that day-to-day price changes are typically small, though several noticeable outliers in both directions. The distribution of log-transformed quantity sold (Figure 2) displays a broad right-skewed pattern with a peak around a log value of four, corresponding to roughly 50–100 sales per day, while also revealing a sizable portion of observations with very low sales volumes.

Figure 1 Daily Log Returns

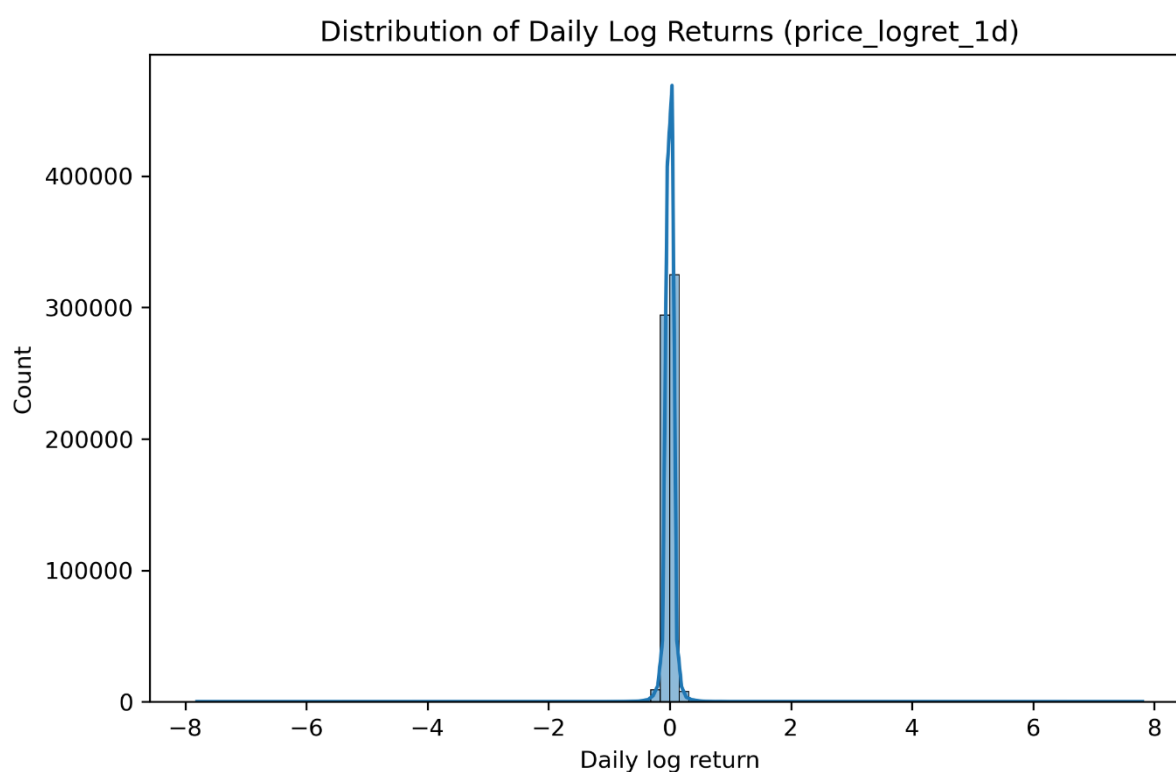
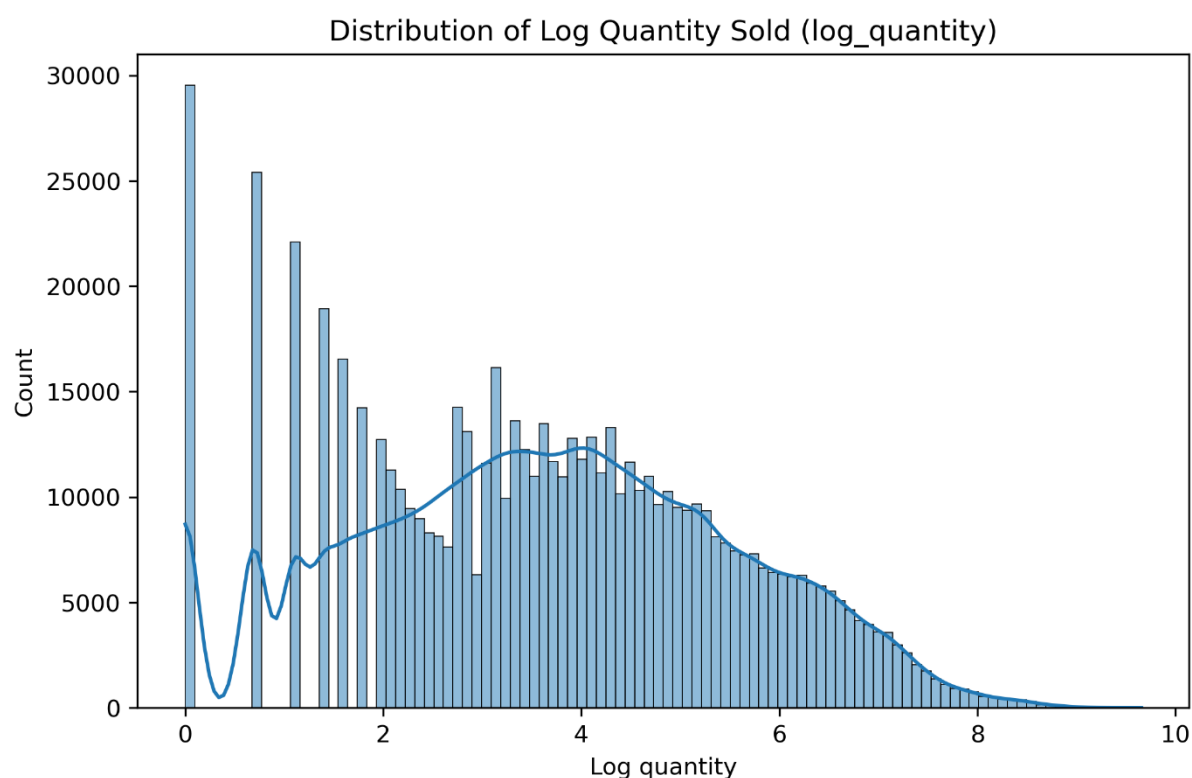
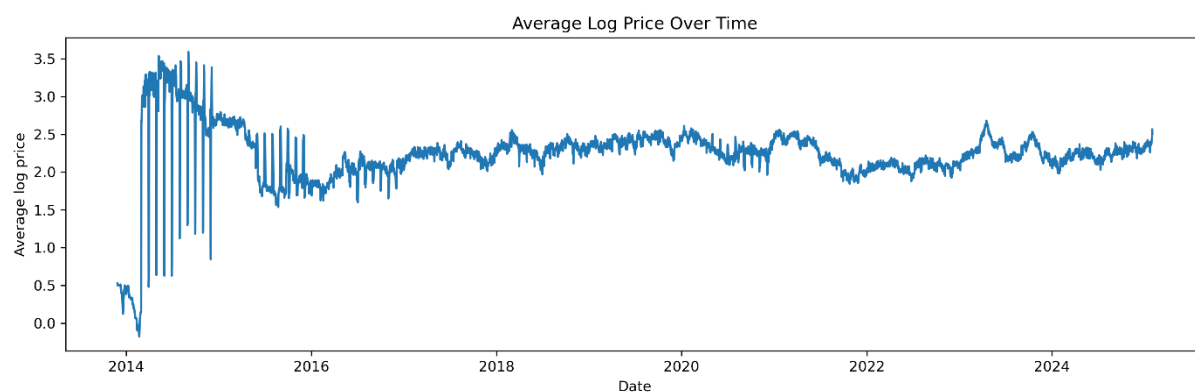


Figure 2 Log quantity sold distribution



The time-series of average log price (Figure 3) illustrates higher volatility during the early years. This is expected because fewer skins existed in 2013–2014, so daily averages were based on a much smaller subset of items. As more skins entered the market, the aggregated series became progressively more stable.

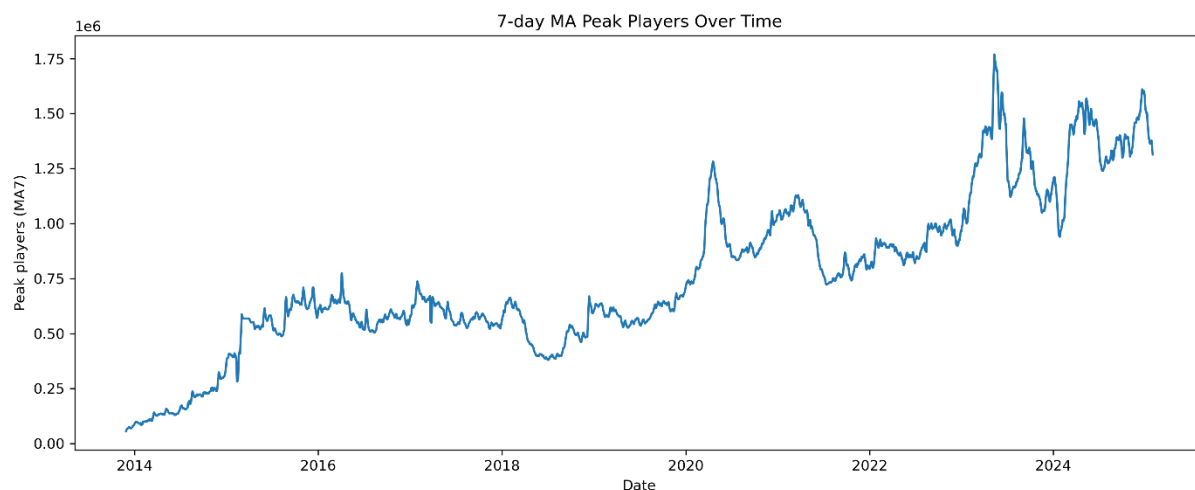
Figure 3 Average log price over time



The seven-day moving average of peak players (Figure 4) shows how peak players has changed over time. There is a clear upwards trend during the CS:GO period from 2013 to 2023. The

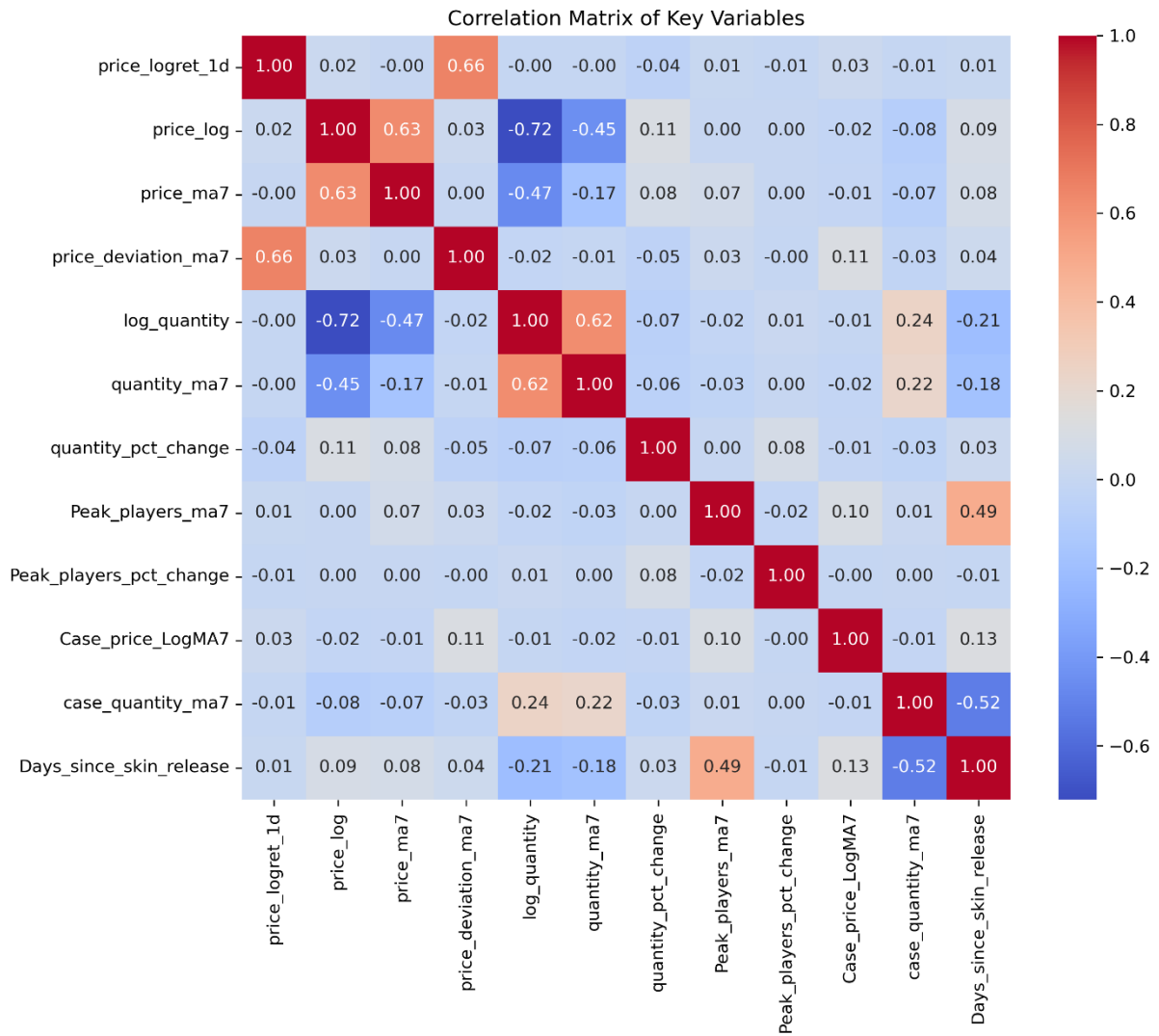
release of CS2 is clearly seen in this figure, where the highest peak is reached in 2023. It is also visible from this that peak player numbers have been more volatile during the CS2 period post 2023.

Figure 4 Peak player 7-day moving average



Finally, the correlation matrix (Figure 5) shows mostly low linear correlations between predictors and the daily log return, suggesting that price changes are driven by short-lived and potentially nonlinear dynamics. Stronger correlations appear among related variables such as price level and its moving average, or log quantity and its moving average, reflecting internal consistency within the engineered features. While some higher correlations exist, this indicates that multicollinearity is not a major issue.

Figure 5 Correlation matrix



5.2. Linear Regression

The split used for training the linear regression was 80/20 and no extra specifications were done for the dataset or features.

The linear regression model achieved an MAE of 0.0280, RMSE of 0.0632 and an R^2 of 0.421. This indicates that a substantial portion of short-term price variation in the CS2 skin market can be explained using linear combinations of the engineered features.

Although daily returns in financial markets are often close to unpredictable, the structured nature of digital item markets, combined with features such as trading volume, case activity, player counts, and moving averages, appears to provide meaningful predictive signals. These results position linear regression as a strong baseline and motivate the use of more flexible nonlinear models to potentially capture additional variance.

The linear regression coefficients (Table 3) highlight several important relationships in the short-term price dynamics of CS2 skins. The strongest predictor by a large margin is price deviation from the 7-day moving average, which has a substantial positive coefficient (0.82). This suggests strong short-term momentum: when a skin trades above its recent trend, the next day's price change is more likely to be positive.

In contrast, the 7-day cumulative return has a negative coefficient (-0.046), indicating mean reversion at the medium-term horizon. Together, these effects imply that price dynamics exhibit short-term continuation but medium-term reversal.

Supply-side case variables also play a role. The log-smoothed case price has a sizable negative effect (-0.157), meaning that increases in case prices are associated with downward pressure on skin returns, likely reflecting increased case-opening activity and corresponding short-term increases in skin supply. Case-release days likewise have a small negative effect (-0.011), while `week_of_case_release` has a positive effect (0.009). This could indicate that players sell skins to open new cases or buy skins from the newly released case, while price then returns over the week.

Demand-side features such as log quantity sold and price tier have small positive coefficients, while increases in daily peak players show a slight negative association with returns. Event variables (Steam Sale, Major, S-tier tournament) display small but directionally consistent effects.

Overall, the coefficient structure suggests that market microstructure variables and moving-average-based momentum indicators play the central role in explaining short-term returns, while item characteristics and calendar effects have relatively minor influence.

Table 3 Linear Regression top and bottom 10 coefficients

Feature	Coefficient
Price_deviation_ma7	0.8223
Week_of_case_release	0.0094
Price_tier_luxury	0.0061
Gun_Glock-18	0.0038
Steamsale_ongoing	0.0029
Gun_M4A4	0.0028
Classification_Mil-Spec Grade	0.0025
Gun_M4A1-S	0.0022

Gun_Desert Eagle	0.0021
Gun_USP-S	0.0018
Condition_MW	-0.0025
Major_ongoing	-0.0031
Weekday_Wednesday	-0.0035
Weekday_Monday	-0.0035
Peak_player_log	-0.0044
Weekday_Sunday	-0.0052
Case_release	-0.0106
Peak_players_pct_change	-0.0123
Price_logret_7d	-0.0465
Case_price_LogMA7	-0.1566

5.3. Random Forest

To capture nonlinear relationships and interactions between variables, a Random Forest Regressor was estimated using the same feature set and training split as the linear model.

The baseline model was trained using the following parameters:

- n_estimators = 200
- max_depth = None
- min_samples_leaf = 5
- max_features = “sqrt”

With these parameters the model achieved:

- Training performance:
 - o MAE: 0.0172
 - o RMSE: 0.0381
 - o R²: 0.7738
- Test performance:
 - o MAE: 0.0278
 - o RMSE: 0.0591
 - o R²: 0.494

The Random Forest model outperformed the linear regression baseline. The model achieved an out-of-sample R^2 of 0.494, compared with 0.421 for linear regression, representing a meaningful improvement of approximately 17%. The test MAE (0.0278) and RMSE (0.0591) indicate that prediction errors remain small in magnitude relative to the typical daily variation in log returns.

The high training R^2 (0.7738) indicates that the model captures complex, nonlinear interactions within the data; however, the gap between training and test performance also reflects a moderate degree of overfitting, which is expected for tree-based ensemble models.

Overall, the Random Forest results suggest that short-term price movements in the CS2 skin market exhibit nonlinear behaviour, and that machine learning methods can extract substantially more predictive signal than linear approaches.

5.3.1. Hyperparameter tuning

To investigate whether predictive performance could be improved and to reduce overfitting, a hyperparameter search was performed using RandomizedSearchCV on a subsample of 100,000 training observations. A three-fold time-series cross-validation scheme was used to preserve chronological order. The searched variables highlighted in Table 4.

Table 4 Random Forest hyperparameter tuning search space

Variable	Search space
n_estimators	[100, 200, 300]
max_depth	[10, 15, 20, None]
min_samples_leaf	[1, 3, 5, 10]
max_features	["sqrt", "log2", 0.3]

The best-performing configuration identified by the search was:

- n_estimators = 200
- max_depth = None
- min_samples_leaf = 5
- max_features = 0.3

This is relatively close to the baseline models parameters, with only max_features being different.

A final model with these parameters achieved a test R^2 of 0.484, slightly lower than the baseline value of 0.494. The MAE (0.0276) and RMSE (0.0597) remained nearly identical to the initial model. These results indicate that hyperparameter tuning does not materially improve predictive performance and that the Random Forest is relatively stable across parameter choices. The similarity between the two models suggests that predictive accuracy is constrained by inherent noise in short-term skin price movements, and that further gains from tuning are unlikely.

5.3.2. *Random Forest feature coefficient*

The Random Forest model provides insight into which predictors contribute most to next-day log returns. The results (Table 5) indicate a highly skewed importance distribution, dominated by variables capturing short-term price dynamics. The most influential predictor was `price_deviation_ma7` (0.639), which measures how far the current price deviates from its seven day moving average. This confirms that short-term mean reversion or momentum effects are the primary drivers of daily price movements in the skin market.

The next most important features were `price_logret_7d` (0.062), reflecting weekly momentum, and `quantity_ma7` (0.032), indicating that sustained trading activity has predictive value. Additional contributors include sudden changes in trading volume (`quantity_pct_change`), fluctuations in player activity (`Peak_players_pct_change`), and a set of supply-side variables such as `Case_price_LogMA7` and `case_quantity_log`. Collectively, these results show that demand dynamics, liquidity, and general market activity meaningfully influence short horizon returns.

In contrast, many variables exhibited near-zero importance, including weekday indicators and most event-related variables (e.g., S-tier tournaments and Majors). This suggests that such events have little direct influence on next-day price changes. The overall pattern resembles financial markets, where high-frequency returns are primarily driven by recent price behaviour and trading intensity, while exogenous events rarely produce consistent short-term predictive signals.

Table 5 Random Forest top 10 coefficients

Feature	Coefficient
<code>Price_deviation_ma7</code>	0.6391
<code>Price_logret_7d</code>	0.0618
<code>Quantity_ma7</code>	0.0322

Quantity_pct_change	0.0266
Peak_players_pct_change	0.0226
Days_since_case_release	0.0207
Days_since_skin_release	0.0205
Case_price_logMA7	0.0199
Case_quantity_log	0.0177
Case_quantity_ma7	0.0176

5.4. XGBoost

To complement the linear regression and Random Forest models, an Extreme Gradient Boosting regressor was estimated to evaluate whether gradient-boosted decision trees could further improve the prediction of short-term price changes. XGBoost often performs well on high-dimensional, noisy tabular data and is designed to capture complex nonlinear interactions that linear models cannot represent.

The model was trained on the same feature set and the same 80/20 train–test split used for the previous models. Tree-based boosting models do not require feature scaling, and all dummy variables and engineered features were included without modification.

The baseline XGBoost model was defined as:

- `n_estimators = 500`
- `learning_rate = 0.05`
- `max_depth = 6`
- `subsample = 0.8`
- `colsample_bytree = 0.8`

These are commonly used conservative defaults that balance flexibility with generalization.

The model was trained using early stopping logic based on test-set performance, due to the older XGBoost version used in the environment.

The XGBoost model achieved the following results:

- Training performance:
 - MAE: 0.0276
 - RMSE: 0.0467
 - R^2 : 0.6599
- Test performance:
 - MAE: 0.0286

- RMSE: 0.0617
- R^2 : 0.4482

Same as with Random Forest, the model performed better on the training data with a R^2 of 0.6599, which is substantially better than the testing data at 0.4482. This indicates a moderate amount of overfitting. The performance of the XGBoost model places it between the linear regression model (0.421) and Random Forest (0.494). This further suggests that the CS2 skin market contains nonlinear patterns that boosted trees can find. However, Random Forest still shows slightly better predictive performance for this dataset.

5.4.1. Hyperparameter tuning

A limited manual tuning procedure was performed by testing multiple combinations of tree depth, learning rate and number of estimators. Results can be seen in Table 6.

This configuration achieved a test R^2 of 0.469, which is only a marginal improvement compared to the baseline (0.448). Some configurations performed worse, and none exceeded the Random Forest baseline.

Given the small improvement and the added computational complexity, the tuned version was not selected as the main model, but results are documented.

Table 6 XGBoost tuning

Estimators	Max Depth	Learning Rate	R^2
300	4	0.05	0.4664
400	6	0.05	0.4482
500	6	0.03	0.4691
600	8	0.03	0.4381

5.4.2. XGBoost feature coefficients

As with the Random Forest, price_deviation_ma7 is by far the most influential feature. This reinforces the conclusion that short-term return predictability in the CS2 skin market is driven primarily by short-horizon mean-reversion or overreaction effects. XGBoost placed more weight on price level features (price_log, price_ma7, median_price), which indicates that boosted trees may be capturing nonlinear relationships between price level and short-term returns.

Table 7 XGBoost top 10 coefficients

Feature	Coefficient
Price_deviation_ma7	0.1726
Price_tier_High	0.0530
Median_price	0.0530
Gun_AWP	0.0466
Pirce_log	0.0442
Price_ma7	0.0326
Weekday_Wednesday	0.0285
Log_quantity	0.0277
Gun_Desert Eagle	0.0247
Gun_USP-S	0.0228

5.5. LSTM

Unlike the other models in this study, which treat each day as an independent observation, an LSTM requires ordered sequences sorted by skin. Therefore, a separate dataset structure was created specifically for the LSTM.

Because LSTMs are designed to learn temporal patterns rather than cross-sectional differences between items, several predictors used in the machine-learning models were excluded. Categorical variables such as weapon type, skin rarity, condition, weekday, and price tier were removed. LSTMs are not well-suited for large sets of static indicators and including them would contribute noise rather than predictive signal. Instead, only numeric time-varying variables were retained.

Since skin are separated by their condition and StatTrak status, these together with name of the skin were used to create ItemID. The dataset could then be separated by the ItemID and ordered by date to create training sets.

A 14-day sliding window was used to construct sequences, with the model predicting next-day price_logret_1d. Sequences were constructed separately for each skin to preserve temporal consistency and avoid mixing unrelated price histories.

The LSTM specification was:

- LSTM layer (64 units, return_sequences = True)
- Dropout layer (20%)

- LSTM layer (32 units)
- Dropout layer (20%)
- Dense output layer (1 unit)

The model was trained using mean squared error (MSE) with the Adam optimizer. A validation split with early stopping was used to prevent unnecessary overfitting. The training set and test set followed the same 80/20 time-based split as the other models.

Model performance was:

- Training performance:
 - MAE: 0.034
 - RMSE: 0.063
 - R^2 : 0.384
- Test performance:
 - MAE: 0.044
 - RMSE: 0.076
 - R^2 : 0.176

The LSTM explains only 17% of the variance in out-of-sample daily log returns, far below the other models in this study. No further tuning was done, since it was unlikely that significant performance could be achieved.

Because LSTMs are complex nonlinear models with many internal parameters, they do not yield interpretable coefficients comparable to those of linear models or tree-based models. Given the relatively modest predictive performance of the LSTM, interpretability analysis was not pursued, and the model is evaluated only in terms of its predictive accuracy.

The LSTM performs worse than the linear regression, Random Forest, and XGBoost models. This suggests that the CS2 skin market does not exhibit strong sequential dependencies that an LSTM can exploit.

5.6. Quantity analysis

Lastly, to evaluate whether major CS2 events affect market activity, log daily transaction volume was analysed using simple linear regression and event–non-event means comparisons.

A linear regression of log volume on event indicators and basic controls explained only 0.5% of the variation in trading activity ($R^2 = 0.0054$), confirming that volume is highly noisy. Several event coefficients were economically meaningful (Table 8). Operations (0.27), case

release weeks (0.26), and increases in peak player activity (0.33) were associated with higher trading volume, while Majors exhibited a small negative association (−0.04).

Table 8 Top and bottom three quantity coefficients

Feature	Coefficient
Peak_players_pct_change	0.3322
Operation_ongoing	0.2769
Week_of_case_release	0.2615
Major_ongoing	-0.0444
Price_logret_1d	-0.0676
Case_realse	-0.1651

Event-based difference-in-means tests provided clearer evidence (Table 9). Trading volume increased substantially during Operations (33.5%) and in the week following a case release (34%). The case-release day itself still showed a strong increase (16.6%). Steam sales produced only a small increase (2.5%), while S-tier tournaments had a modest effect (8.1%). In contrast, trading activity decreased slightly during Majors (−8%), suggesting a temporary shift of attention away from the marketplace.

Table 9 Event-based differences in quantity sold

Event	N (event=1)	Mean log qty (1)	Mean log qty (0)	Difference	% Difference
Steam sale	66457	3.6549	3.6302	0.0247	2.5%
Major	34900	3.5545	3.6373	-0.0828	-7.9%
S-tier	273797	3.6773	3.5995	0.0777	8.1%
Operation	145170	3.8564	3.5672	0.2892	33.5%
Case Week	28203	3.9148	3.6198	0.2950	34%
Case Day	3961	3.7857	3.6318	0.1539	16.6%

5.7. Results summary

To compare the predictive performance of all models, Table 9 summarizes their MAE, RMSE, and R^2 values on the test dataset. The Random Forest model achieved the strongest

performance, with an out-of-sample R^2 of 0.494, outperforming both the linear regression baseline ($R^2 = 0.421$) and XGBoost ($R^2 = 0.448$). All tree-based models consistently delivered lower prediction error than the linear model, confirming the presence of nonlinear relationships in short-term CS2 skin price movements.

The LSTM model performed significantly worse than all other methods, with an R^2 of only 0.176, indicating that the dataset contains limited exploitable sequential structure. Overall, the results suggest that traditional machine-learning models, particularly ensemble tree methods, are well-suited for predicting short-term CS2 skin price changes, while deep learning architectures offer little additional value for this task.

Table 10 Model performance comparison

Model	MAE	RMSE	R^2
Linear Regression	0.028023	0.063224	0.421179
Random Forest	0.027818	0.059111	0.494044
XGBoost	0.028574	0.061733	0.448154
LSTM	0.043734	0.075448	0.176154

6. Discussion

The purpose of this study was to evaluate whether short-term price movements in the CS2 skin market can be predicted using a combination of economic, behavioural, and event-based features. The results show that Random Forest and XGBoost outperformed Linear Regression, while the deep learning model LSTM performed considerably worse than all of them.

The low performance of the LSTM model suggests that there is little meaningful memory in past price series. Meaning that past prices only help in predicting future values to a small degree, and temporal dependencies are weak in the skin's economy. The Random Forest model delivered the highest performance, suggesting that there are some nonlinear relationships and interactions between the features.

Events had little to no effect on price, while they did impact sales quantities to a small degree. This could indicate that supply is generally pretty good and that even if demand rises, it does not impact the price. It is quite understandable that some events, like Steam sales do not impact pricing or quantities sold since player might not be interested in other games, even at discounted prices. Majors being the only event with less sales during them is interesting. This

could reflect that people engage less with the game and the systems around it while the tournament is happening.

LSTMs had in previous research achieved good results, especially within stock price predictions, while in this study it performed poorly. One possible reason for this is simply that the CS2 skin's economy lacks strong temporal dependencies. It is also possible that the structure of this dataset is not enough for LSTMs to find these dependencies. With individual sales and a better structured dataset, the performance of a deep learning model like LSTM could improve significantly. The 640,000 observations are on the lower side, compared to previous research where the number of observations are in the millions for many studies.

6.1. What does the study say about the skin's economy

As previous research has suggested, the prices of skins depend on scarcity and social aspects. This highlighted by scholars like Werning (2019) when comparing cosmetics in game to fashion and Thorhauge (2023) mentioning that the value of skins comes from the prestige of owning them. Yamamoto and McArthur (2015) also highlighted design, as well as supply and demand elements as important role in skins prices. They also note that this market is in fully control of the developers, an idea like Lehdonvirta and Ernkqvist (2011) when they called scarcity in virtual 41 economies artificially created. These factors explain why one skin can be valued at one US dollar, while another is hundreds or thousands of US dollars. The potential to use trade-up contracts to exchange skins to a higher rarity plays a key role in keeping prices of certain skins higher than they otherwise would.

While these factors help explain long-run price levels, the aim of this study was to examine short-term price dynamics. The results indicate that internal and external events have only a limited effect on daily price movements. Instead, the strongest predictors are price inherent features, such as the seven-day moving average and deviation from trend. This suggests that short-term movements are driven primarily by internal market momentum rather than reactions to external events.

The study further shows that supply-side case variables and player activity indicators contribute modestly to predictability, whereas esports events, Steam sales, and operations have negligible influence on daily returns. This aligns with the notion that short-term price changes in cosmetic markets are dominated by trader behaviour, liquidity fluctuations, and micro-level demand rather than major structural events. Moreover, the relatively low explanatory ceiling (R^2 of around 0.50 for the best model) suggests that a substantial share of price movement is noisy, unpredictable, or driven by factors not captured in the dataset.

Taken together, the evidence suggests that the CS2 skin market behaves more like a behavioural market with short-lived momentum and substantial random variation, rather than an efficient market where prices fully reflect available information. This is also not that unexpected as previous research suggested, since skins are not purely financial vehicles.

6.2. Data limitations

One of the main limitations of the study is that structure of the original data through the Steam API. Not having access to individual sales limits the study and modelling performance. While median price means outliers are automatically removed, it does not show the full picture of the market. This also means that price spikes and hidden micro patterns could be hidden behind median price.

Both wear rating and pattern template are missing from the data as well, and while it would be ideal to have access to these, they do not impact the study as much as the lack of individual sales. This is partly because of the chosen skins not having differences in pattern template. For skins sold higher or lower compared to average due to their wear rating, the use of median price might eliminate some of these outliers. Together with individual sales it would be interesting to have access to information such as, how long a skin has been on the market before being sold. This could give more insight into how the market work and allow more in-depth modelling.

Further on the study only used 28 skins out of over 1,300 skins, and while the skin's used do represent some of the more popular skins, it does mean that the results of this study cannot generalize the whole market. Other gun types and ways of acquiring skins could result in different market patterns. Especially knives and gloves, which are the rarest items, could behave in a unique way, which this study cannot comment on.

Some missing values are also present within the data, while relatively small (7,808), it does mean that there is a lack of continuity within the data and calculations like moving average. This may have caused some smoothing that alter some patterns, making it harder for the models to detect them.

Because of the data used is over eleven years long it does present some issues. The reason for using such a long period was to simply model the whole existence of skins, as well as mitigate some of the issues caused by lack of individual reports. This together with events happening less infrequent in the CS2 skins market compared to stocks or other financial assets, it was deemed reasonable to use a longer time period. However, this means that major happenings have occurred under this period. The game has become a lot more popular over

time, the game being released in China, updates to trading (e.g. Steam Community Market limits), and the release of CS2. These have undoubtedly influenced the market and made it less stable for some periods; these shifts introduce non-stationarity, making short-term predictive modelling more difficult.

6.3. Methodological limitations

The use of a chronological train–test split is appropriate for forecasting, but it also reflects structural changes in the dataset. New skins appear gradually, meaning the early training period is dominated by fewer items. Some events (especially operations) occurred frequently early on but rarely in recent years, resulting in an uneven distribution of event data.

All models in this study predict one-day-ahead log returns. While this is standard for financial-style forecasting, longer horizons (e.g., 7-day or 30-day returns) may reveal more persistent patterns. Moving averages help in noise reduction and improve stability but may remove some volatility. The choice of a 7-day window was arbitrary, and another range could have been implemented.

The absence of established theory for which variables drive skin price movements meant that feature engineering was necessarily broad. However, a more systematic feature reduction process (e.g., recursive feature elimination, mutual information screening, or regularization methods) may have produced a more better performing and potentially more stable model.

Most existing research on price prediction focuses on financial assets such as equities or cryptocurrencies, where high-frequency data, well-established modelling practices, and decades of theory (e.g., market microstructure; return predictability) provide a strong foundation for methodological choices. In contrast, the CS2 skin market has not been studied in a systematic, data-driven way, and there is no established framework for feature engineering, event modelling, or handling the irregular structure of supply and demand in virtual goods.

As a result, some modelling decisions in this thesis were exploratory in nature and guided by analogy to financial research rather than by existing domain-specific literature. This does not weaken the validity of the analysis but highlights the limited state of research in digital cosmetic markets and the need for methodological development in future studies.

6.4. Future research

Future work on CS2 skin price prediction would benefit substantially from improved data availability and methodological extensions. The most important limitation in the present study

is the absence of individual transaction-level data. Access to full sales histories, including exact sale price, wear rating, pattern template, listing duration, and time on market, would enable modelling of market effects that are currently hidden behind median price aggregates. Such data would allow deep learning models such as LSTMs to find temporal dependencies not found in this study.

Expanding the scope of analysed items would also strengthen external validity. This thesis focused on 28 commonly traded rifle and pistol skins, but the broader CS2 economy includes over 1,300 cosmetic items. Knives and gloves represent a distinct subset of the market. Modelling these items separately could reveal dynamics that do not appear in more liquid, lower-tier skins.

More advanced modelling approaches may also provide additional insight. Longer horizon predictions (e.g., 7-day or 30-day returns) could help identify structural drivers of price movement beyond short-term momentum.

Overall, future research should combine improved micro-level data with more specialised modelling techniques to better capture the behavioural and structural dynamics of the CS2 skin market.

7. Conclusion

This thesis examined whether short-term price movements in the CS2 skin market can be predicted using machine-learning models and a range of events and market-structure features. The results demonstrate that while some predictive signal exists, short-term returns remain difficult to forecast with high accuracy. Random Forest and XGBoost models provided the strongest performance, achieving out-of-sample R^2 values close to 0.50, while Linear Regression performed slightly worse and LSTM showed the weakest predictive ability. This suggests that short-term pricing dynamics in the skin market are nonlinear but largely driven by noise and micro-level fluctuations that are not captured by available data.

The analysis indicates that internal price-based features, such as recent moving averages and short-term deviations from trend, are far more influential than external factors. Tournaments, Steam sales, and operations had minimal impact on daily returns, though some effects were observable in sales quantities. Supply-related variables, such as case prices and quantities sold, contributed modestly to prediction accuracy but were not dominant drivers.

Taken together, the findings suggest that the CS2 skin market exhibits characteristics of a behavioural and trader-driven economy with limited short-term efficiency. Momentum appears to play a role, but a substantial portion of price changes remains unexplained. Future

research using individual transaction data, broader item coverage, and richer temporal features may uncover deeper structure within this emerging digital market.

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Appendix

Appendix A – Raw data

A1. List of skins

Weapon	Skin	Rarity	Case	Year
AK-47	Elite Build	Mil-Spec Grade	Chroma 2 Case	2015
AK-47	Redline	Classified	Operation Phoenix Weapon Case	2014
AK-47	Slate	Restricted	Snakebite Case	2021
AK-47	Vulcan	Covert	Huntsman Weapon Collection	2014
AWP	Asiimov	Covert	Operation Phoenix Weapon Case	2014
AWP	Atheris	Restricted	Prisma Weapon Case	2019
AWP	Capillary	Mil-Spec Grade	Prisma 2 Case	2020
AWP	Fever Dream	Classified	Spectrum Case	2020
Desert Eagle	Crimson Web	Restricted	eSports 2014 Summer Case	2014
Desert Eagle	Mecha Industries	Classified	Danger Zone Case	2018
Desert Eagle	Printstream	Covert	Fracture Case	2020
Desert Eagle	Blue Ply	Mil-Spec Grade	Prisma 2 Case	2020
Glock- 18	Bunsen Burner	Mil-Spec Grade	Falchion Case	2015
Glock- 18	Vogue	Classified	Fracture Case	2020
Glock- 18	Wasteland Rebel	Covert	Gamma Case	2016
Glock- 18	Water Elemental	Classified	Operation Breakout Weapon Case	2014
M4A1-S	Blood Tiger	Mil-Spec Grade	CS:GO Weapon Case	2013
M4A1-S	Chantico's Fire	Covert	Chroma 3 Case	2016

M4A1-S	Night Terror	Restricted	Dreams & Nightmares Case	2022
M4A1-S	Printstream	Covert	Operation Broken Fang Case	2020
M4A4	Cyber Security	Classified	Operation Broken Fang Case	2020
M4A4	Magnesium	Mil-Spec Grade	Danger Zone Case	2018
M4A4	Neo-Noir	Covert	Clutch Case	2018
M4A4	The Emperor	Covert	Prisma Case	2019
USP-S	Black Lotus	Mil-Spec Grade	Operation Riptide Case	2021
USP-S	Cyrex	Restricted	Glove Case	2016
USP-S	Kill Confirmed	Covert	Shadow Case	2015
USP-S	Printstream	Covert	Recoil Case	2022

A2. Data gathering

Primary data was gathered through the Steam API. The search requires being logged into a steam account and using a URL to target certain skins (e.g., AK-47 Redline would look like: https://steamcommunity.com/market/pricehistory/?appid=730&market_hash_name=AK-47%20%7C%20Redline%20%28Field-Tested%29). This provides a JSON object containing timestamp, median sales price, quantity sold.

Peak player data was scraped from their publicly available chart showing peak players for each day. This data contained date and peak player count.

Appendix B – Feature definition

Variable	Type	Description
StatTrak	Int(0/1)	1 if skin is StatTrak
Date	Datetime	Date of observation
Median_price	Float	Median sales price
Quantity_sold	Int	Quantity sold
Days_since_skin_release	Int	Observation date minus skin release date

Case_release	Int	1 if case was released that day
Week_of_case_release	Int	1 if case release was within a week
Days_since_case_release	Float	Counter for days since case was last case release
Operation_ongoing	Int(0/1)	1 if operation is ongoing
Major_ongoing	Int(0/1)	1 if major is ongoing
Days_since_major_end	Float	Counter for days since last major
S_tier_ongoing	Int(0/1)	1 if S-tier tournament is ongoing (Majors removed)
Steamsale_ongoing	Int(0/1)	1 if steam sale is ongoing
price_ma7	Float	7-day moving average of price
price_log	Float	Log-transformed price
price_logret_1d	Float	Daily log return (TARGET)
price_logret_7d	Float	7-day log return
price_deviation_ma7	Float	Deviation of today's log price from 7-day moving average
quantity_ma7	Float	7-day moving average of quantity
log_quantity	Float	Log-transformed quantity
quantity_pct_change	Float	Daily percentage change in quantity
Case_price_LogMA7	Float	7-day moving average of Log-transformed case price
case_quantity_log	Float	Log-transformed case quantity
case_quantity_ma7	Float	7-day moving average of case quantity
Peak_players_log	Float	Log-transformed peak players
Peak_players_pct_change	Float	Daily percentage change in peak palyers
Peak_players_ma7	Float	7-moving average of peak players
Condition*	Int(0/1)	Condition dummy variables
Gun*	Int(0/1)	Gun dummy variables
Classification*	Int(0/1)	Classification dummy variables
Weekday*	Int(0/1)	Weekday dummy variables
Price_tier*	Int(0/1)	Price tier dummy variables

Several variables were log-transformed to stabilise variance and reduce skewness in highly volatile raw prices and quantities. Moving averages (7-day windows) were included to capture short-term trends, and logarithmic returns were used to express relative changes in price in a scale-independent manner.

Dropped features include:

- Type: gun type (e.g., rifle, pistol)
- Release_year: skin release year
- Skin: name of skin
- Skin_key: feature used to join case price and quantity to correct skin and date
- Skin_case: feature used to join case price and quantity to correct skin and date
- Peak_players: raw number of peak players
- Case_price: raw number of case price
- Case_quantity: raw number of case quantity
- S_tier_prize_pool_usd: the prize pool of S_tier_ongoing in USD

Appendix C – Model Specifications

Model	Key parameters
Random Forest (baseline)	n_estimators=200, max_depth=None, min_samples_leaf=5, n_jobs=-1, random_state=42
XGBoost (baseline)	n_estimators=500, learning_rate=0.05, max_depth=6, subsample=0.8, colsample_bytree=0.8, objective="reg:squarederror", n_jobs=-1, random_state=42
LSTM	seq_len = 14, LSTM Layer 1: 64 units, Dropout: 0.2, LSTM Layer 2: 32 units, Dropout: 0.2, Dense Output Layer: 1 unit, validation_split=0.1, epochs=30, batch_size=256,

All models done with 80/20 training-testing split.

Random Forest tuning search space:

Variable	Search space
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n_estimators	[100, 200, 300]
max_depth	[10, 15, 20, None]
min_samples_leaf	[1, 3, 5, 10]
max_features	["sqrt", "log2", 0.3]

Random Forest tuning ran with sample_size = 100_000, RandomizedSearchCV, and n_iter=20.

XGBoost tuning search space:

Config	Key parameters
1	{"n_estimators": 300, "max_depth": 4, "learning_rate": 0.05}
2	{"n_estimators": 500, "max_depth": 6, "learning_rate": 0.05}
3	{"n_estimators": 400, "max_depth": 6, "learning_rate": 0.03}
4	{"n_estimators": 600, "max_depth": 8, "learning_rate": 0.05}

Appendix D – Additional tables

Full list of Linear Regression coefficients

Feature	Coefficient
price_deviation_ma7	0.82234109
Week_of_case_release	0.00939108
price_tier_Luxury	0.0061184
Gun_Glock-18	0.00375459
Steamsale_ongoing	0.00297892
price_tier_High	0.0028057
Gun_M4A4	0.0027687
Classification_Mil-Spec Grade	0.00254612
Gun_M4A1-S	0.00221156
Gun_Desert Eagle	0.00212183
Gun_USP-S	0.0018066
price_log	0.00172968
log_quantity	0.00158142
Classification_Restricted	0.00158021

price_tier_Mid-range	0.00120922
StatTrak	0.00104971
Median_price	0.00085035
Gun_AWP	0.00080753
case_quantity_log	0.00012393
S_tier_ongoing	8.8851E-05
Condition_WW	6.645E-05
Days_since_last_case_release	6.4292E-06
Days_since_major_end	3.6322E-06
Quantity_sold	2.6419E-06
Days_since_skin_release	1.864E-07
case_quantity_ma7	5.1585E-09
Peak_players_ma7	2.779E-09
quantity_ma7	-3E-06
Weekday_Saturday	-0.0007234
price_ma7	-0.0008664
Operation_ongoing	-0.0009989
Classification_Covert	-0.0011583
Weekday_Thursday	-0.0018443
Condition_FN	-0.0018605
quantity_pct_change	-0.0022405
Weekday_Tuesday	-0.0022789
Condition_FT	-0.0024301
Condition_MW	-0.0024708
Major_ongoing	-0.003136
Weekday_Wednesday	-0.0035209
Weekday_Monday	-0.0035313
Peak_players_log	-0.0044439
Weekday_Sunday	-0.0051596
Case_release	-0.0105783
Peak_players_pct_change	-0.0122664
price_logret_7d	-0.0464852
Case_price_LogMA7	-0.1566001

Random Forest looked at features (top 20).

Feature	Importance
price_deviation_ma7	0.639084
price_logret_7d	0.061767
quantity_ma7	0.032236
quantity_pct_change	0.026592
Peak_players_pct_change	0.022640
Days_since_last_case_release	0.020746
Days_since_skin_release	0.020514
Case_price_LogMA7	0.019900
case_quantity_log	0.017671
case_quantity_ma7	0.017609
Days_since_major_end	0.017561
Peak_players_log	0.014862
Peak_players_ma7	0.014476
Quantity_sold	0.011247
log_quantity	0.010426
price_ma7	0.008963
Median_price	0.008064
price_log	0.007964
Weekday_Wednesday	0.002461
S_tier_ongoing	0.002312

XGBoost looked at features (top 20).

Feature	Importance
price_deviation_ma7	0.172589
price_tier_High	0.053044
Median_price	0.053010
Gun_AWP	0.046601
price_log	0.044155
price_ma7	0.032624
Weekday_Wednesday	0.028509

log_quantity	0.027702
Gun_Desert Eagle	0.024758
Gun_USP-S	0.022865
price_logret_7d	0.022809
Quantity_sold	0.022223
quantity_ma7	0.021551
Weekday_Monday	0.019854
Weekday_Tuesday	0.019171
quantity_pct_change	0.019081
Classification_Covert	0.019041
Week_of_case_release	0.018640
StatTrak	0.018366
Condition_MW	0.018319